

PROJECT DESIGN PHASE : SOLUTION ARCHITECTURE DOCUMENT

SZ METRO TICKET GENERATING SYSTEM IN SERVICENOW

Date: 31 July 2026

Team / Dev ID: 23JR1A05E4 (Individual)

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Max Marks: 4 Marks

SYSTEM ARCHITECTURE DIAGRAM & DATA FLOW

LAYER / SUBSYSTEM	SERVICENOW / TECHNOLOGY COMPONENT & FUNCTIONAL RESPONSIBILITY
Presentation Tier (Client UI)	ServiceNow Service Portal, AngularJS, HTML5, CSS3, JS Canvas: Provides responsive mobile ticket ordering UI, dynamic dropdown filters for station selection, live fare preview rendering, and on-screen scannable QR generation for turnstiles.
Application Logic Tier (Server Side)	Script Includes (GlideAjax), Business Rules, Flow Designer Engine: • Fare Calculator: Executes distance matrix lookups dynamically on station selection. • Lifecycle Controller: Automatically triggers status transitions (Active → Used / Expired). • Notification Service: Dispatches automated SMS/Email receipts via Flow Designer.
Data Storage Tier (Database)	ServiceNow Relational Engine (MariaDB backed): Houses transit schema tables: u_metro_ticket (ticket logs), u_station_matrix (fare grid), sys_user (profiles), and audit trail logs (sys_audit).
Integration & Security Tier	REST APIs, Webhooks, Access Control Lists (ACLs), RBAC: Enforces strict data security based on user roles (u_metro_passenger, u_metro_inspector) and manages payment API callbacks.

KEY ARCHITECTURAL OBJECTIVES & QUALITY ATTRIBUTES

System Goals & Non-Functional Requirements:

- **Low Latency Response:** Asynchronous GlideAjax client-server interaction guarantees total booking and QR ticket generation in under 1.5 seconds.
- **High Availability & Cloud Elasticity:** Hosted on ServiceNow enterprise PaaS cloud, capable of handling peak morning rush hour transaction bursts.
- **Security & Fraud Prevention:** Dynamic, encrypted payload QR generation combined with real-time status invalidation prevents ticket reuse or duplication.

Document Version: 1.0.0 | Project: SZ Metro Ticket Generating System in ServiceNow | Developer: Shaik Zuveriya Tabassum (23JR1A05E4)